STALKS OF THE PUSHFORWARD ALONG A CLOSED IMMERSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for both cases of the displayed formula.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter etale-cohomology.

Editorial note. The preamble is the chapter's own native preamble excerpt already used for this source: the Xy-pic setup (native preamble lines 14-18) and the native theorem declarations with the mathematical macros (native preamble lines 154-217).

Allowed standard prerequisites (editorial).

- Closed immersion i : Z -> X and the open complement U = X minus Z. Same chapter, the open immersion of the complement and the fact that the small etale site of X is covered by Z and U. Supplies the two cases of the proof: a geometric point inside Z and one outside.
- Stalk of a sheaf at a geometric point and the description of the pushforward by sections. Same chapter, the stalk functor at a geometric point and the description of sections of i_{small,*}G. Supplies the computation in both cases.

Source reference. Original lemma 04FW: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6621–6621. Navigation pointer invoked at proof line 6693; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04FV: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6597–6597. Navigation pointer invoked at proof line 6694; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Date: stacks-project:etale-cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04E6: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6696–6696. Navigation pointer invoked at proof line 6696; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 04E7: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 6698–6698. Navigation pointer invoked at proof line 6698; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Lemma 0.1. *Let $i : Z \to X$ be a closed immersion of schemes. Let $\mathcal{G}$ be a sheaf of sets on $Z_{étale}$. Let $\overline{x}$ be a geometric point of X. Then*

$$(i_{small,*}\mathcal{G})_{\overline{x}} = \begin{cases} * & \text{if } \overline{x} \notin Z \\ \mathcal{G}_{\overline{x}} & \text{if } \overline{x} \in Z \end{cases}$$

where $*$ denotes a singleton set.

Proof. Let $U = X \setminus Z$. Note that $i_{small,*}\mathcal{G}|_{U_{étale}} = *$ is the final object in the category of étale sheaves on U, i.e., the sheaf which associates a singleton set to each scheme étale over U. This explains the value of $(i_{small,*}\mathcal{G})_{\overline{x}}$ if $\overline{x} \notin Z$.

Next, suppose that $\overline{x} \in Z$. Note that

$$(i_{small,*}\mathcal{G})_{\overline{x}} = \operatorname{colim}_{(U,\overline{u})} \mathcal{G}(U_Z)$$

and on the other hand

$$\mathcal{G}_{\overline{x}} = \operatorname{colim}_{(V,\overline{v})} \mathcal{G}(V).$$

Let $\mathcal{C}_1 = \{(U, \overline{u})\}^{opp}$ be the opposite of the category of étale neighbourhoods of $\overline{x}$ in X, and let $\mathcal{C}_2 = \{(V, \overline{v})\}^{opp}$ be the opposite of the category of étale neighbourhoods of $\overline{x}$ in Z. The canonical map

$$\mathcal{G}_{\overline{x}} \longrightarrow (i_{small,*}\mathcal{G})_{\overline{x}}$$

corresponds to the functor $F : \mathcal{C}_1 \to \mathcal{C}_2$, $F(U, \overline{u}) = (U_Z, \overline{x})$. Now Lemmas 04FW and 04FV imply that $\mathcal{C}_1$ is cofinal in $\mathcal{C}_2$, see Categories, Definition 04E6. Hence it follows that the displayed arrow is an isomorphism, see Categories, Lemma 04E7. □